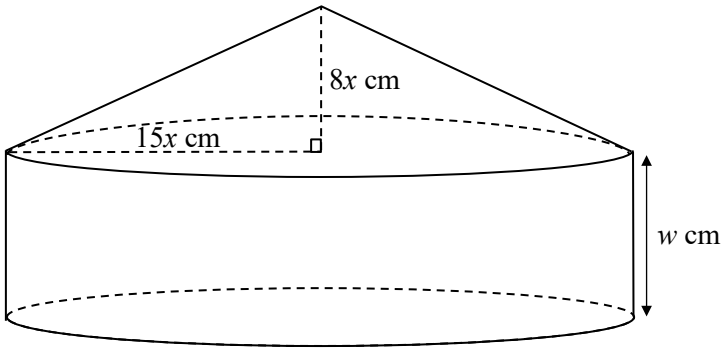
RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 1 (Lambda)
Topic(s): Calculus I
Differentiation, Applications, Maclaurin and Graphing (Questions)

1 [It is given that a right circular cone with base radius r and height h has volume $\frac{1}{3}\pi r^2 h$ and curved surface area πrl, where l is the slanted height of the cone.]

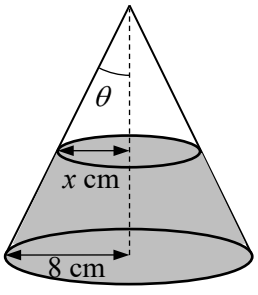


The model of a house is made up of three parts.

- The roof is modelled by the curved surface of a right circular cone with base radius $15x$ cm and height $8x$ cm.
- The walls are modelled by the curved surface of a cylinder of radius $15x$ cm and height w cm.
- The floor is modelled by a circular disc of radius $15x$ cm.

The three parts are joined together as shown in the diagram. The model is made of material of negligible thickness. It is given that the volume of the model is a fixed value k cm^3. Use differentiation to find the exact values of x and w in terms of k such that the external surface area is a minimum. Simplify your answers. [8]

2



Water is poured at a rate of 8 cm^3 per second into a conical container with base radius 8 cm. The semi-vertical angle of the cone is θ, where $\tan\theta = 0.4$. At time t seconds after the start, the radius of the water surface is x cm (see diagram). Find the rate of increase of the depth of water when the depth is 10 cm.

[The volume of a cone of base radius r and height h is given by $V = \frac{1}{3}\pi r^2 h$.] [7]

3 A curve C has equation $y^2 + xy = -1$, where $y > -1$.

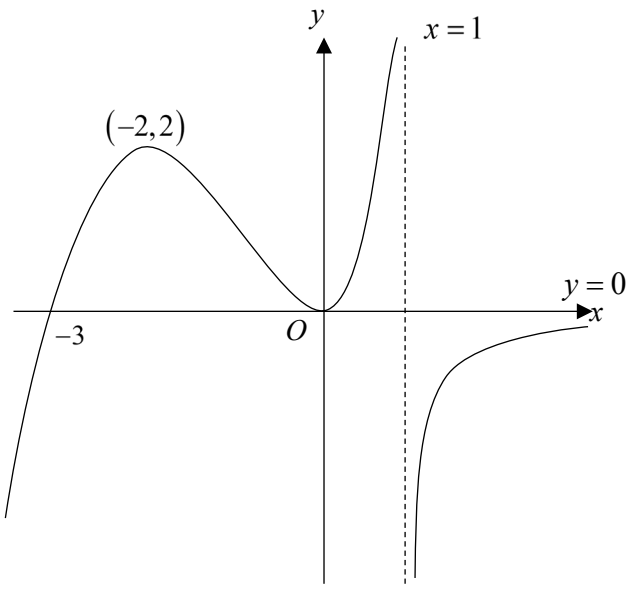
(a) Describe a sequence of transformations that will transform the graph of C onto the graph of $4y^2 - 2(x-1)y = -1$. [3]

(b) **(i)** Show that the gradient function of C can be expressed as $\dfrac{dy}{dx} = \dfrac{-y}{2y+x}$. [2]

(ii) For a non-zero constant a, find the equation of the tangent to the point where $y = a$ in terms of a. [3]

(iii) State the behaviour of the tangent as $a \to 1$. [1]

4 **(a)** The graph of $y = \mathrm{f}(x)$ is given below. It has one vertical asymptote at $x = 1$ and one horizontal asymptote at $y = 0$. The graph cuts the x-axis at $x = -3$ and has turning points at $(-2, 2)$ and the origin.



Sketch the graph of $y = \mathrm{f}'(x)$, stating the equations of any asymptotes, and indicating any axial intercepts. [3]

(b) It is given that $\mathrm{g}(x) = x^3 - 3x + p$, where p is an unknown constant.

(i) Given that the graph of $y = \dfrac{1}{\mathrm{g}(x)}$ has a vertical asymptote at $x = 2$, find the value of p. [1]

(ii) Given instead that the graph of $y = \dfrac{1}{\mathrm{g}(x)}$ has a minimum point at $y = \dfrac{1}{5}$, find the value of p. [3]

5 A curve is defined parametrically by

$$x = \frac{at}{t+1}, \; y = \frac{at^2}{t+1},$$

where a is a non-zero constant and $t \in \mathbb{R}, t \neq -1$.

(i) Show that the equation of the normal to the curve at the point T with parameter t is given by $t(t+1)(t+2)y + (t+1)x = at(t^3 + 2t^2 + 1)$. [4]

(ii) The normal to the curve at the point $P\left(-a, \frac{1}{2}a\right)$ meets the curve again at point Q. Find the exact coordinates of Q in terms of a. [4]

(iii) Let M be the mid-point of PQ. Find a cartesian equation of the curve traced by M if a varies. [2]

6 The curve C has equation $y = \frac{x^2 + x + 3}{x - 2}$.

(i) Without using a calculator, find the set of values that y can take. [4]

(ii) Sketch the graph of C, stating clearly the equations of any asymptotes and the coordinates of any turning points and axial intercepts. [3]

(iii) By drawing a suitable graph on the same diagram in **(ii)**, find the range of values of k, where $k > 0$, such that the equation $(x+1)^2(x-2)^2 + \left(x^2 + x + 3\right)^2 = k^2\left(x-2\right)^2$ has at least one positive real root, giving your answer correct to 3 decimal places. [3]

7 **(a)** Expand $\frac{4 + x^2}{\sqrt{4 - x}}$ in ascending powers of x, up to and including the term in x^2. [3]

State the set of values of x for which the expansion is valid. [1]

(b) It is given that $y = \ln(1 + \sin x)$.

(i) Show that $\left(1 + \sin x\right)\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \left(\cos x\right)\frac{\mathrm{d}y}{\mathrm{d}x} = -\sin x$. [2]

(ii) By further differentiation of the result in part **(i)**, find the Maclaurin series for y, up to and including the term in x^3. [4]

(iii) Hence deduce the Maclaurin series for $\ln\left(\frac{1 - \sin x}{2}\right)$, up to and including the term in x^3. [2]

8 **(a)** In the triangle ABC, $AC = BC = 1$ and angle $ACB = \left(\frac{1}{2}\pi + \theta\right)$ radians. Given that θ is sufficiently small angle, show that

$$AB \approx \sqrt{2} + \frac{\sqrt{2}}{2}\theta - \frac{\sqrt{2}}{8}\theta^2$$ [5]

(b) Let $\mathrm{f}(x) = \ln(2 - x)$.

(i) Write down the first three non-zero terms of the Maclaurin series for $\mathrm{f}(x)$, where $-2 \le x < 2$.

[3]

Denote the answer to part **(i)** by $\mathrm{g}(x)$.

(ii) Find the range of values of x for which the percentage error between the value of $\mathrm{g}(x)$ and the value of $\mathrm{f}(x)$ is within 10%. [2]